Rushyanth Nerellakunta

(317) 935-0710 | Indianapolis, IN, USA | rushyanth.work@gmail.com | LinkedIn-nerellakunta-rushyanth | GitHub-RushyanthN

Professional Summary

Machine Learning Engineer with **3+ years** of experience building **production-grade** intelligent systems, from **voice-driven** browser interfaces to **RAG-based** Q&A pipelines. Proven track record of architecting **scalable GenAI solutions** on **AWS**, improving **SQL generation accuracy** from **78% to 92%**, and reducing **data pipeline latency by 40%**. Expert in **LLM orchestration**, **multi-agent systems**, and **MLOps**, with hands-on experience deploying models using **PyTorch**, **LangChain**, and **cloud platforms**. Strong collaborator skilled at translating complex AI capabilities into customer-facing products and actionable business insights.

Professional Experience

- **Kahana Inc (Remote)** — Indianapolis, USA
  AI/ML Engineer — *Aug 2025 - Present*
  - Built a **voice-first browsing layer** on top of **Firefox**, enabling users to talk to the browser for navigation and task completion (shopping, job applications, search) instead of manual clicking/typing.
  - Implemented an **AWS Lambda**–hosted **GenAI gateway** to securely run **Deepgram** (speech-to-text) and **Gemini** (intent + reasoning) while isolating API credentials from the client and preventing key exposure.
  - Added **semantic + request-level caching** (e.g., **Redis**) to reduce duplicate LLM calls, mitigate API rate-limit risk, and stabilize performance under repeated/looping user queries during load tests.
  - Created and tested a **multi-agent orchestration system** using **Google ADK**, decomposing workflows into specialized sub-agents (e.g., Shopping, Job Apply, Q&A/Search) with routing + handoffs for multi-step tasks.
  - Implemented **structured reasoning + tool-use guardrails** (**ReAct-style** planning/execution) with retrieval/tool validation to reduce off-target responses and **hallucination risk** for long, complex voice context.
  - Integrated **MCP (Model Context Protocol)** to standardize tool connectivity and context exchange between the browser layer, agents, and external services—improving extensibility and faster onboarding of new tools.
  - Partnered with product + engineering to translate research prototypes into **customer-facing AI capabilities**, iterating through test feedback, usability constraints, and reliability requirements.
- **Tech Stack:** Python, AWS Lambda, FastAPI, WebSockets, Deepgram API, Gemini, Redis, Google ADK, MCP (Model Context Protocol), Docker, AWS S3, CloudWatch, Cognito
- **COMET Lab, Indiana University** — Indianapolis, IN
  Machine Learning Engineer — *Aug 2024 - May 2025*
  - Built an **end-to-end Natural Language → SQL** Q&A system by cleaning web-scraped CSV datasets, loading them into a **SQL database**, and using **LangChain + Llama-3.3-70B-Versatile** to generate SQL, execute and return results to users.
  - Implemented **embedding-based retrieval** and stored vectors in a local **ChromaDB** to fetch relevant context (e.g., schema snippets, table/column descriptions) before generation—improving grounding and reducing ambiguous-field errors.
  - Implemented **schema-aware RAG** by injecting relevant tables/columns + schema context into prompts, improving overall SQL generation performance and reducing ambiguous-field failures.
  - Improved **SQL execution accuracy from 78% → 92%** through iterative **prompt engineering** (one-shot/few-shot exemplars) and **structured planning + tool-use prompting** (**ReAct-style**) for complex joins and multi-step analytical questions.
  - Added **read-only guardrails** (validation + blocklist/allowlist) to prevent destructive statements (DELETE/UPDATE/DROP) and ensure safe execution for critical queries; integrated **Human-in-the-Loop review** for low-confidence or high-impact requests.
  - Built a **post-training evaluation harness** with **60+ ground-truth test cases**, reporting **Execution Accuracy** and **Exact Match**, and used error analysis to systematically iterate on prompts, schema context, and safety checks.
  - Presented research findings and recommendations to technical and non-technical stakeholders, influencing strategic roadmap decisions.
- **Tech Stack:** Python, SQL, LangChain, ChromaDB, Embeddings, Llama-3.3-70B-Versatile, Local LLMs, RAG (schema/context retrieval), Evaluation (Exact Match, Execution Accuracy, semantic similarity), Guardrails, HITL
- **Virgenverse** — Hyderabad, India
  Jr Machine Learning Engineer — *Feb 2022 - Nov 2023*
  - Engineered resilient **ETL pipelines** using **Apache Airflow + Python**, reducing **data refresh latency by 40%** and ensuring T+1 data availability to support downstream ML training and reporting.
  - Built a **sentiment analytics pipeline** with **spaCy/NLTK** to process **100K+ social/customer posts**, generating actionable insights and dashboards for marketing teams.
  - Designed and executed **A/B and multivariate experiments** to optimize pricing and marketing initiatives, improving conversion rates.
  - Optimized data warehousing in **Snowflake** by migrating legacy stored procedures into efficient SQL transformations, improving **BI/dashboard query performance by 3×** and reducing operational complexity.
  - Improved **data quality and reliability** with validation checks, retry logic, and lineage-friendly logging, reducing pipeline failures and speeding up root-cause analysis.

- Collaborated with data engineers, product managers, and marketing teams to translate insights into actionable recommendations.
- **Tech Stack:** Python, Apache Airflow, SQL, Snowflake, spaCy, NLTK, BI Dashboards

Projects

- **Modular RAG with Agentic Routing and Self-Correction for Financial Document Analysis** Dec 2025
  GitHub: RushyanthN/Advanced-Modular-RAG-with-Agentic-Routing-and-Self-Correction
  - Built a production-grade RAG pipeline over Apple and Tesla SEC 10-K filings, processing **3,751 financial documents** (**2,016 pages + 1,735 tables**) and enabling natural-language Q&A with source-level citations across multi-year reports
  - Implemented hierarchical chunking with **6,068 parent chunks** (2,048 chars) and **20,300 child chunks** (512 chars), improving retrieval recall while reducing context fragmentation by **40%** and achieving **95%** answer accuracy on complex financial queries.
  - Developed LLM-driven query expansion using RAG Fusion and step-back prompting, generating multiple semantic variants per query and fusing results via **Reciprocal Rank Fusion**, improving retrieval recall by **35%** while reducing irrelevant results by **50%**.
  - Applied a two-stage retrieval pipeline with bi-encoder semantic search (**BGE-base-en-v1.5**) followed by cross-encoder re-ranking (**BGE-reranker-large**) and diversity filtering, achieving **0.92+** relevance scores and improving answer completeness by **30%**.
  - Built a confidence-aware generation layer with adaptive response strategies, automated citation extraction, and audit-ready metadata attribution (company, year, page), maintaining **98%** response appropriateness across diverse financial query types.
- **Tech Stack:** Python, ChromaDB (HNSW), Sentence-Transformers (BGE-base-en-v1.5, BGE-reranker-large), LangChain, Ollama (Llama 3.2), PyTorch (CUDA), PyMuPDF/fitz, pdfplumber, pandas, NumPy

- **SparkFlow — Streamlined Machine Learning with Apache Spark** May 2025
  GitHub: RushyanthN/Machine-Learning-with-Apache-Spark
  - Built an **end-to-end distributed ML pipeline** in **Apache Spark ML** implementing **Linear Regression, Logistic Regression, K-Means, and LDA**, with automated JSON metrics logging for reproducible experiment tracking.
  - Trained a **Linear Regression model** on a **692-dimensional feature space** (80/20 split), achieving **R^2 = 0.73, RMSE = 0.26, and MSE = 0.068**.
  - Developed a binary **Logistic Regression classifier** on 692-dimensional sparse vectors, achieving **100% accuracy / precision / recall / F1** with **AUC-ROC = 1.0** on the test set.
  - Implemented **K-Means** with k comparison (k=2 vs k=3), improving **WSSSE from 0.12 → 0.075 (37.5% better)** and generating automated cluster distribution summaries.
  - Built an **LDA topic modeling pipeline** extracting **10 topics**, comparing hyperparameters across k=[5,10], and reporting perplexity (k=5: 2.78, k=10: 3.10) plus log-likelihood for model selection.
- **Tech Stack:** Python, Apache Spark, PySpark (Spark MLlib), Spark SQL, argparse (CLI), JSON metrics logging, Jupyter Notebook, Pandas/NumPy (EDA), Git/GitHub

Skills

Programming Skills: C, C++, Python, R, SQL, Bash, UNIX.

Machine Learning: Supervised & Unsupervised Learning, Deep Learning (CNN, RNN, Transformers), NLP, Recommendation Systems, PyTorch, TensorFlow, Hugging Face Transformers.

Data Analytics: Exploratory Data Analysis (EDA), Feature Engineering, Statistical Modeling, Hypothesis Testing, A/B Testing, Tableau, Power BI, Matplotlib, Seaborn, Plotly.

Big Data & Cloud: Apache Spark, Hadoop, AWS SageMaker, SQL & NoSQL Databases, ETL Pipelines, Data Warehousing (Snowflake, Redshift), Azure ML, GCP Vertex AI.

MLOps & Deployment: FastAPI, Docker, Kubernetes, CI/CD, MLflow, Weights & Biases.

Tools & Version Control: MATLAB, MixPanel, Git, GitHub, Cuda, Jenkins. Visual Studio, JIRA.

Education

- **Indiana University Purdue University Indianapolis** Indianapolis, IN
  Master of Science, Applied Data Science; GPA: 3.8/4.0 *May 2025*
- **Sreenidhi Institute of Science and Technology** Hyderabad, India
  Bachelor of Technology, Electronics & Communication Engineering *May 2022*

Certifications

- **AWS Machine Learning Engineer – Associate** Valid Jul 2025 - Jul 2028